PBL-3 · 27 OCTOBER 2026

WORKING IMPLEMENTATION · PRELIMINARY EVIDENCE

Reliability-Aware Audio-Visual Fusion for Generalizable Deepfake Detection

A compact detector that estimates how much to trust each modality in a short video clip.

Aditya Sahu · RA2311026010814
Akash Cherian · RA2311026010852
Prachurya Subhasmita · RA2311026010855

Report Writing · 21CSC402P · SRMIST

01 · RESEARCH PROBLEM

Audio and video do not always deserve equal trust

Challenge

- Manipulation may affect the face, voice, or both.
- Compression, blur, noise, and missing faces can weaken one stream.
- Synchronization mismatch is informative but is not a fake/real verdict.

Project question

Does a lightweight reliability estimator improve detection over unimodal and equal-weight fusion baselines, especially when modality quality differs?

The gate is evaluated empirically; its benefit is not assumed.

Two compact branches feed one reliability-aware decision

Visual + audio

16 face frames · MobileNetV3
16 kHz log-Mel · DS-CNN
quality features · (p_v, p_a)

→

Reliability gate

quality · uncertainty
synchronization
weights (w_v,w_a)

→

Fused classifier

weighted features
fused probability
(p_f)

2-second window · 16 sampled frames · 16 kHz audio · (w_v+w_a=1)

The gate uses both branches. Synchronization remains an auxiliary input, not an independent verdict.

Implementation uses an OpenCV Haar face detector; landmark alignment remains future work.

Leakage-aware manifests are in place; the primary test stays locked

PARTITION	CLIPS	REAL	FAKE	USE
Train	15,018	350	14,668	Fit
Validation	2,136	50	2,086	Model selection and threshold
Test	4,390	100	4,290	Reserved; not evaluated
Total	**21,544**	**500**	**21,044**	FakeAVCeleb v1.2

Stack

Python · PyTorch/torchvision · OpenCV · torchaudio · scikit-learn/SciPy · YAML

Demo

Local browser frontend returns a fused score, modality scores, reliability weights, and quality flags.

About 97.7% of each split is fake; training uses balanced sampling. Licensed clips stay local.

04 · PRIMARY PAIRED AV SNAPSHOT

Equal-weight fusion leads after the first validation epoch



What this says

Fixed fusion is +0.0801 AUROC over audio-only. Reliability fusion is −0.1469 below fixed fusion.

One epoch, one seed, validation only. No uncertainty interval; no final test metrics.

Metric: ROC-AUC · higher is better

05 · EXTERNAL AUDIO-ONLY TRANSFER

The FakeAVCeleb audio branch did not transfer to this probe

0.3777

AUROC · grouped-bootstrap 95% CI [0.3588, 0.3964]

Full German MLAAD-tiny slice · 2,820 windows · 1,491 utterance groups

Fixed source threshold

0.295, selected on FakeAVCeleb validation; not retuned on the target.

Balanced accuracy 0.4827
Recall 0.8820 · specificity 0.0833

Interpretation: target labels mean bona-fide vs. spoof audio; training used FakeAVCeleb clip_fake. Report as a separate exploratory transfer failure, not as paired AV performance.

Negative findings are retained and reported.

The implementation works; the reliability hypothesis remains open

Current evidence does not show that reliability-aware fusion improves generalization. The primary test split remains untouched.

Now	Train longer; repeat seeds; inspect gate weights and modality losses.
Then	Run degradation, missing-modality, and held-out manipulation evaluation.
Final	Select and calibrate on validation; freeze the checkpoint; evaluate test once.

Local demo

Show the clip score, (p_v), (p_a), (w_v), (w_a), and quality flags.

State clearly: the current checkpoint is a one-epoch development artifact; a detector score is not proof of manipulation.

```
make web CHECKPOINT=artifacts/checkpoints/reliability_fusion/best_model.pt
```